

Metric Spaces II: Lecture Notes

Transcribed from Handwritten Notes

1 Completeness and Continuity

Theorem 1.1. *Let (X, d) be a metric space such that every Cauchy sequence in X has a convergent subsequence. Then (X, d) is complete.*

Proof. Let $\{x_n\}$ be a Cauchy sequence, and let $\{x_{n_k}\}$ be a subsequence of $\{x_n\}$ that converges to x .

Claim 1. $x_n \rightarrow x$.

Let $\epsilon > 0$. Since $\{x_n\}$ is Cauchy, there exists N such that if $m, n \geq N$, then:

$$d(x_m, x_n) < \frac{\epsilon}{2}$$

Since $x_{n_k} \rightarrow x$, there is a natural number k such that $n_k > N$ and $d(x_{n_k}, x) < \frac{\epsilon}{2}$.

Then, for any $n \geq N$:

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Thus, $x_n \rightarrow x$. □

Remark 1.1. If d is the usual metric on $\mathbb{R}$, then $(\mathbb{R}, d)$ is complete.

1.1 Continuous Functions

Theorem 1.2. *Let $(X, \mathcal{T})$ and $(Y, \mathcal{U})$ be topological spaces and let $f : X \rightarrow Y$. Then the following statements are equivalent:*

(a) f is continuous.

(b) If $V \in \mathcal{U}$, then $f^{-1}(V) \in \mathcal{T}$ (Inverse image of open is open).

(c) If C is a closed subset of $(Y, \mathcal{U})$, then $f^{-1}(C)$ is a closed subset of $(X, \mathcal{T})$.

Proof of (b) $\Rightarrow$ (c). Let C be a closed subset of $(Y, \mathcal{U})$. Then $Y - C \in \mathcal{U}$. By (b), $f^{-1}(Y - C) \in \mathcal{T}$. Now, note that $f^{-1}(C) = X - f^{-1}(Y - C)$. Therefore, $f^{-1}(C)$ is the complement of an open set in X , making it a closed subset of $(X, \mathcal{T})$. □

2 Compactness

Theorem 2.1 (Bolzano-Weierstrass). *Every infinite bounded subset of $\mathbb{R}$ has a limit point. Equivalently, every bounded sequence in $\mathbb{R}$ has a convergent subsequence. A subset of $\mathbb{R}^n$ is sequentially compact iff it is closed and bounded.*

Theorem 2.2 (Heine-Borel). *Let $[a, b]$ with $a < b$ and let $\mathcal{O}$ be a collection of open intervals such that*

$$[a, b] \subseteq \bigcup_{O \in \mathcal{O}} O.$$

Then there is a finite subset $\{O_1, O_2, \dots, O_N\}$ of $\mathcal{O}$ such that

$$[a, b] \subseteq \bigcup_{n=1}^N O_n.$$

Definition 2.1 (Cover). A collection $\mathcal{A}$ of subsets of a topological space $(X, \mathcal{T})$ is a **cover** of a subset B of X provided $B \subseteq \bigcup \mathcal{A}$. If each member of $\mathcal{A}$ is open (in $\mathcal{T}$), it is called an **open cover** of B .

Definition 2.2 (Compactness). A subset A of a topological space $(X, \mathcal{T})$ is **compact** provided every open cover of A has a finite subcover.

2.1 Examples of Compactness

Example 2.1. If X is a finite set and $\mathcal{T}$ is any topology on X , then $(X, \mathcal{T})$ is compact.

Example 2.2. Let $A = [a, b]$. Let $\mathcal{U}$ be an open cover of the closed interval $[a, b]$. For each $x \in [a, b]$, there exists $U_x \in \mathcal{U}$ such that $x \in U_x$. Since U_x is open, there is an open interval I_x such that $x \in I_x \subseteq U_x$. By the Heine-Borel Theorem, there is a finite sub-collection $\{I_{x_1}, \dots, I_{x_n}\}$ such that $[a, b] \subseteq \bigcup I_{x_i}$. The collection $\{U_{x_1}, U_{x_2}, \dots, U_{x_n}\}$ of members of $\mathcal{U}$ that corresponds to the intervals is a finite sub-collection of $\mathcal{U}$ that covers $[a, b]$. Therefore $[a, b]$ is compact.

Example 2.3. The open interval $(0, 1)$ is not compact because the collection

$$\left\{ \left(\frac{1}{n}, 1 \right) : n \in \mathbb{N} \right\}$$

is an open cover that does not have a finite sub-cover.

Example 2.4. $\mathbb{R}$ is not compact because $\{(-n, n) : n \in \mathbb{N}\}$ is an open cover that does not have a finite sub-cover.

2.2 Totally Bounded Metric Spaces

Definition 2.3 (ϵ -net). Let (X, d) be a metric space and let $\epsilon > 0$. A finite subset F of X is called an **ϵ -net** for X if for each $x \in X$, there exists $p \in F$ such that $d(x, p) < \epsilon$.

Definition 2.4 (Totally Bounded). A metric space (X, d) is **totally bounded** if for each $\epsilon > 0$, there is an ϵ -net for X .

Theorem 2.3. *Every totally bounded metric space is bounded.*

Proof. Let (X, d) be a totally bounded metric space. Let F be a 1-net for X . Let $M = \max\{d(p, q) : p, q \in F\}$. Let $x, y \in X$. Then there exist $a, b \in F$ such that $d(x, a) < 1$ and $d(y, b) < 1$. Therefore,

$$d(x, y) \leq d(x, a) + d(a, b) + d(b, y) < 1 + M + 1 = M + 2.$$

Hence, $d(x, y) < M + 2$ for all $x, y \in X$. Thus, (X, d) is bounded. $\square$

Example 2.5. For $x, y \in \mathbb{R}$, let $d(x, y) = \min\{|x - y|, 1\}$. Then $(\mathbb{R}, d)$ is bounded, since $d(x, y) \leq 1$ for all x, y . However, it is **not** totally bounded. There is no $\frac{1}{2}$ -net for $\mathbb{R}$. Let F be any finite subset of $\mathbb{R}$ and let p be the largest member of F . If $q \in F$, then $d(p + 1, q) = 1$ (since distance is capped at 1 or actual distance > 1), hence there is no point in F within distance $1/2$ of $p + 1$.

Theorem 2.4. *A metric space (X, d) is compact if and only if it is complete and totally bounded.*

Definition 2.5 (Sequential Compactness). A topological space $(X, \mathcal{T})$ is sequentially compact provided every sequence in X has a subsequence that converges.

3 Uniform Continuity

Definition 3.1. Let (X, d) and (Y, ρ) be metric spaces and let $f : X \rightarrow Y$ be a function. Then f is **uniformly continuous** if for each $\epsilon > 0$, there is a positive number δ such that if $x, y \in X$ and $d(x, y) < \delta$, then $\rho(f(x), f(y)) < \epsilon$.

Theorem 3.1. *Let (X, d) be a compact metric space, and let (Y, ρ) be a metric space, and let $f : X \rightarrow Y$ be a continuous function. Then f is uniformly continuous.*

Theorem 3.2 (Continuous Image of Compact Space). *The continuous image of a compact space is compact.*

Proof. Let $(X, \mathcal{T})$ be a compact space, let $(Y, \mathcal{U})$ be a topological space, and let $f : X \rightarrow Y$ be a continuous function that maps X onto Y . Let $\mathcal{O}$ be an open cover of Y . Then $\mathcal{O}' = \{f^{-1}(V) : V \in \mathcal{O}\}$ is an open cover of X . Since X is compact, $\mathcal{O}'$ has a finite sub-cover, say $\mathcal{O}''$. The set $\{V \in \mathcal{O} : f^{-1}(V) \in \mathcal{O}''\}$ is a finite sub-collection of $\mathcal{O}$ that covers Y . Therefore, Y is compact. $\square$

3.1 Proof: Continuous on $[a, b] \Rightarrow$ Uniformly Continuous

Theorem 3.3. *If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then f is uniformly continuous.*

Proof. For each $x \in [a, b]$ and given $\epsilon > 0$, there exists $\delta_x > 0$ such that

$$|x - y| < \delta_x \Rightarrow |f(x) - f(y)| < \frac{\epsilon}{2}.$$

The family

$$\left\{ \left(x - \frac{\delta_x}{2}, x + \frac{\delta_x}{2} \right) \right\}_{x \in [a, b]}$$

is an open cover of $[a, b]$. By the compactness of $[a, b]$, we can extract a finite sub-cover:

$$[a, b] \subseteq \bigcup_{i=1}^N \left(x_i - \frac{\delta_{x_i}}{2}, x_i + \frac{\delta_{x_i}}{2} \right).$$

Choose $\delta = \min_{1 \leq i \leq N} \frac{\delta_{x_i}}{2} > 0$. Now, if $|x - y| < \delta$, then x must be in one of the intervals of the finite sub-cover, say around x_i : $|x - x_i| < \frac{\delta_{x_i}}{2}$. By the triangle inequality:

$$|y - x_i| \leq |y - x| + |x - x_i| < \delta + \frac{\delta_{x_i}}{2} \leq \delta_{x_i}.$$

Thus, by the local continuity at x_i :

$$|f(x) - f(y)| \leq |f(x) - f(x_i)| + |f(x_i) - f(y)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Hence f is uniformly continuous. □

4 Connectedness

From an intuitive point of view, we say that something is connected if it consists of one piece. Spaces that are not connected are also interesting. Connectedness was defined for bounded, closed subsets of $\mathbb{R}$ by Cantor in 1883, but his definition is not suitable for general topological spaces. In 1892, Camille Jordan gave a different definition.

Definition 4.1. A topological space is **connected** if it cannot be expressed as the union of two non-empty, disjoint open sets.

Definition 4.2 (Separated Sets). Let $(X, \mathcal{T})$ be a topological space and let A and B be subsets of X . Then A and B are **separated** in X if

$$\overline{A} \cap B = A \cap \overline{B} = \emptyset.$$

If A and B are separated in X and $X = A \cup B$, then A and B are said to form a **separation** of X .

Theorem 4.1. A topological space $(X, \mathcal{T})$ is connected iff it cannot be expressed as the union of two non-empty sets that are separated in X .

Proof. Suppose X is not connected. Then there are non-empty, disjoint open sets U and V such that $X = U \cup V$. Then U and V are also closed (since $U = X - V$ and $V = X - U$). Thus $\overline{U} \cap V = U \cap V = \emptyset$ and $U \cap \overline{V} = U \cap V = \emptyset$. Therefore U and V are separated in X .

Conversely, suppose there are non-empty subsets A and B such that $X = A \cup B$ and $\overline{A} \cap B = \emptyset$ and $A \cap \overline{B} = \emptyset$. Since $X = A \cup B$ and $\overline{A} \cap B = \emptyset \Rightarrow \overline{A} \subseteq A$, so A is closed. The same argument shows that B is closed. Since $A = X \setminus B$ and B is closed, A is open. Similarly B is open. Therefore A and B are both open and closed, and hence X is not connected. □

4.1 Examples of Connectedness

Example 4.1. Let X be a set with more than one member, and let $\mathcal{T}$ be the discrete topology on X . Then $(X, \mathcal{T})$ is not connected (any single point is open and closed).

Example 4.2. Let $X = \mathbb{R} \setminus \{0\}$ with the usual topology. Since $X = (-\infty, 0) \cup (0, \infty)$, $(X, \mathcal{T})$ is not connected.

Example 4.3. Is $X = (\mathbb{R}, \mathcal{T})$ (usual topology) connected or disconnected? Let $\mathcal{T}$ be the usual topology on $\mathbb{R}$. Then $(\mathbb{R}, \mathcal{T})$ is connected.

Proof. Suppose $(\mathbb{R}, \mathcal{T})$ is not connected. Then there exist disjoint non-empty open sets U, V such that $\mathbb{R} = U \cup V$. Since U, V are complements of each other, they are also closed. Let $a \in U$ and $b \in V$. W.L.O.G, assume $a < b$. Let $W = U \cap [a, b]$. Since W is bounded, it has a least upper bound c . Since W is closed (intersection of closed sets), $c \in W \subseteq U$. Since $U \cap V = \emptyset$, $c \neq b$. Since c is the L.U.B of W , $(c, b] \subseteq V$. Therefore $c \in \overline{V}$. But V is closed, so $c \in V$. Therefore $c \in U \cap V$, which is a contradiction. $\square$

Remark 4.1. A subset of $(\mathbb{R}, \mathcal{T})$ is connected iff it is an interval.

4.2 Continuous Image and Applications

Theorem 4.2. Let $(X, \mathcal{T})$ be a connected space, let $(Y, \mathcal{U})$ be a topological space, and let $f : X \rightarrow Y$ be a continuous function that maps X onto Y . Then $(Y, \mathcal{U})$ is connected.

Proof. Suppose $(Y, \mathcal{U})$ is not connected. Then there exist disjoint non-empty open sets U, V such that $Y = U \cup V$. Since f is continuous, $f^{-1}(U)$ and $f^{-1}(V)$ are non-empty open subsets of X . Since $U \cap V = \emptyset$, $f^{-1}(U) \cap f^{-1}(V) = \emptyset$. Also $X = f^{-1}(Y) = f^{-1}(U) \cup f^{-1}(V)$. This implies $(X, \mathcal{T})$ is not connected, a contradiction. $\square$

Corollary 4.3 (Intermediate Value Theorem). Let $f : X \rightarrow \mathbb{R}$ be a continuous function, where X is connected. Let $a, b \in X$ such that $f(a) < f(b)$, and let $d \in \mathbb{R}$ such that $f(a) < d < f(b)$. Then there exists $c \in X$ such that $f(c) = d$.

Proof. $f(X)$ is a connected subset of $\mathbb{R}$. Moreover, $f(X)$ is an interval. Therefore $[f(a), f(b)] \subseteq f(X)$, and so there exists $c \in X$ such that $f(c) = d$. $\square$

Example 4.4 (Topologist's Sine Curve). Let

$$A = \{(0, y) \in \mathbb{R}^2 : -1 \leq y \leq 1\}$$

$$B = \left\{ (x, y) \in \mathbb{R}^2 : 0 < x \leq 1 \text{ and } y = \sin\left(\frac{\pi}{x}\right) \right\}$$

Let $X = A \cup B$. Let $\mathcal{T}$ be the subspace topology induced on X by the usual topology on $\mathbb{R}^2$. Then $(X, \mathcal{T})$ is connected.

Reasoning: Define $f : (0, 1] \rightarrow B$ by $f(x) = (x, \sin(\frac{\pi}{x}))$. Then f is continuous and maps $(0, 1]$ onto B . Since $(0, 1]$ is connected, B is connected. But $\overline{B} = X$ (the closure includes the limit points on the y-axis, A). Since the closure of a connected set is connected, X is connected.

5 Pathwise and Local Connectedness

Definition 5.1 (Path). A **path** in a topological space $(X, \mathcal{T})$ is a continuous function $f : [0, 1] \rightarrow X$. The path f is called a path from $f(0)$ to $f(1)$.

Definition 5.2 (Pathwise Connected). A topological space $(X, \mathcal{T})$ is **pathwise connected** if for each pair $a, b \in X$, there is a path f in X such that $f(0) = a$ and $f(1) = b$.

Theorem 5.1. *Every pathwise connected space is connected.*

Proof. Suppose there is a pathwise connected space $(X, \mathcal{T})$ that is not connected. Then there exist open sets U, V such that $X = U \cup V$ and $U \cap V = \emptyset$. Let $a \in U$ and $b \in V$. Let f be a path from a to b . Then $f^{-1}(U)$ and $f^{-1}(V)$ are non-empty disjoint open subsets of $I = [0, 1]$ such that $I = f^{-1}(U) \cup f^{-1}(V)$. This implies the interval I is disconnected, which is a contradiction. $\square$

Remark 5.1. The Topologist's Sine Curve is connected but **not** pathwise connected. (There is no path connecting a point in A to a point in B).